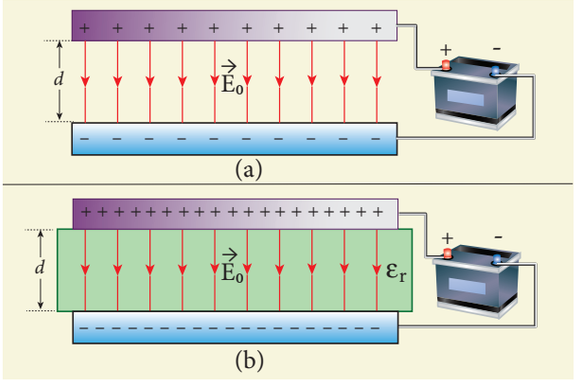



**Figure 1.57** (a) Capacitor is charged through a battery (b) Dielectric is inserted when the battery is connected.

$$Q = \epsilon_r Q_o \qquad (1.97)$$

Due to this increased charge, the capacitance is also increased. The new capacitance is

$$C = \frac{Q}{V_0} = \epsilon_r \frac{Q_0}{V_0} = \epsilon_r C_0 \qquad (1.98)$$

However the reason for the increase in capacitance in this case when the battery remains connected is different from the case when the battery is disconnected before introducing the dielectric.

Now, $C_0 = \dfrac{\epsilon_0 A}{d}$

and $C = \dfrac{\epsilon A}{d}$ (1.99)

The energy stored in the capacitor before the insertion of a dielectric is given by

$$U_0 = \frac{1}{2} C_0 V_0^2 \qquad (1.100)$$

Note that here we have not used the expression $U_0 = \frac{1}{2}\frac{Q_0^2}{C_0}$ because here, both charge and capacitance are changed, whereas in equation (1.100), $V_o$ remains constant.

After the dielectric is inserted, the capacitance is increased; hence the stored energy is also increased.

$$U = \frac{1}{2} C V_0^2 = \frac{1}{2} \epsilon_r C_0 V_0^2 = \epsilon_r U_0 \qquad (1.101)$$

Since $\epsilon_r > 1$ we have $U > U_o$.

It may be noted here that since voltage between the capacitor $V_0$ is constant, the electric field between the plates also remains constant.

The energy density is given by

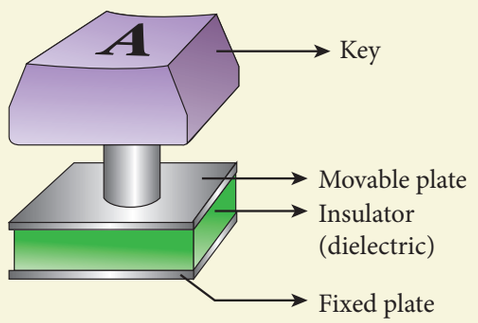
$$u = \frac{1}{2} \epsilon E_0^2 \qquad (1.102)$$

where $\epsilon$ is the permittivity of the given dielectric material.

The results of the above discussions are summarised in the following Table 1.2




Computer keyboard keys are constructed using capacitors with a dielectric as shown in the figure.




When the key is pressed, the separation between the plates decreases leading to an increase in the capacitance. This in turn triggers the electronic circuits in the computer to identify which key is pressed.



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